YTTEK

Academic and Research Solutions

PluSDR Lite

User Manual

YTPC1002C01 Platform Quick Start Guide

For MATLAB Development Environment

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Version： 1.1

tomorrow's wireless, today

Outline

1. INTRODUCTION 4
2. GETTING STARTED 5
3. VERIFYING THE KIT CONTENTS 6
   3.1 KIT CONTENTS 6
   3.2 INSTALLING YTPC1002C01 PLATFORM 7
   3.3 YTPC1002C01 PLATFORM FRONT PANEL 8
   3.4 YTPC1002C01 PLATFORM BACK PANEL 9
4. CONFIGURING THE DEVICE AND COMPUTER ENVIRONMENT SETTING 10
   4.1 REQUIRED SOFTWARE 10
   4.2 ETHERNET INTERFACE 10
   4.3 CONFIGURING THE HOST ETHERNET INTERFACE 11
5. PLATFORM SOFTWARE QUICK START 15
   5.1 STEP1: INITIALIZATION 15
   5.2 STEP2: RUN EXAMPLE 16
6. INTRODUCTION OF MATLAB APIS AND EXAMPLES 17
   6.1 TRANSMIT I/Q RAW DATA : TX() 17
   6.2 RECEIVE I/Q DATA : RX() 19
   6.3 TRX LOOP BACK : TX_RX() 21
      6.3.1 Example 1: Single Tone Loopback Test 21
      6.3.2 Example 2: OFDM Loop Back Test 23
   6.4 OFDM EXAMPLE 24
      6.4.1 Example 3: OFDM EVM Test with Coding (Transfer Pictures) 24

6.4.2 Example 4: MIMO OFDM Test 25

6.5 RF SETTING 28

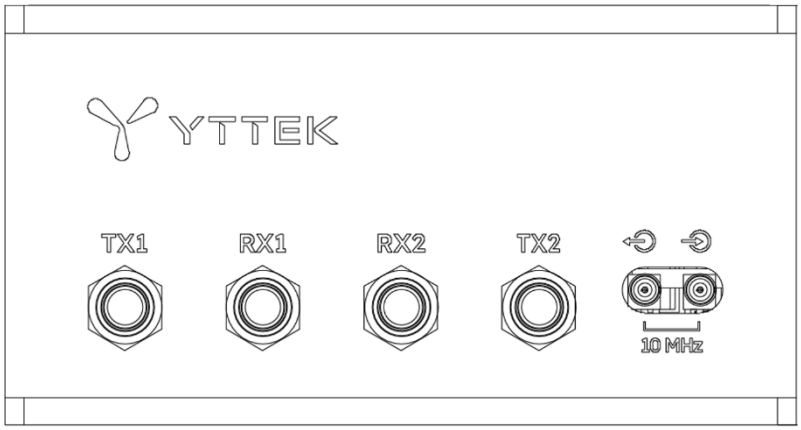
6.5.1 Set Transmitted Power: set_TX_power() 28
6.5.2 Set RX Reference Level: set_RX_Ref_Level_ELSDR() 28
6.5.3 Set Center Frequency : LO_CHANGE() 28

7. TROUBLESHOOTING **29**

8. DOCUMENT VERSION HISTORY **30**

1. Introduction

YTPC1002C01



Features:

RF Capabilities

• Maximum 2 TX, 2 RX

• 300 MHz to 6 GHz frequency band

• Up to 100 MHz bandwidth per channel

Baseband Processing

• Xilinx FPGA

Software

• MATLAB

• C/C++

• Python

Synchronization

• REF IN (10MHz clock reference input)

• REF OUT (10MHz clock reference output)

• TRIG IN

Signal In/Out

• 3.5mm SMA female connectors

• 1 RJ45 (1 GbE)

• 1 Type-C USB to JTAG

Power

• DC 12V, 2A

Form Factor

• 134.71mm x 86mm x 45.62mm

Applications:

- WiFi IEEE802.11b/11a/11n/11ac
- Bluetooth
- 4G LTE TDD/FDD transceiver
- 5G NR sub-6GHz RF transceiver
- NB-IoT

2. Getting Started

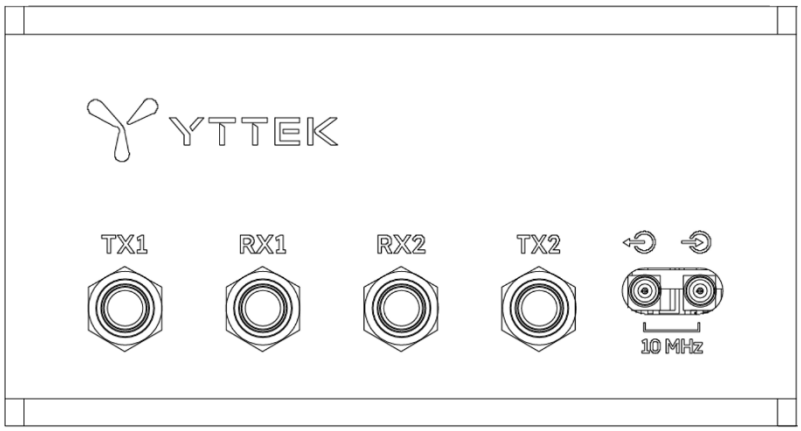
The purpose of this document is to familiarize users with the feaures and usage of YTPC1002C01 platform. Use this document to help users get started with setting up the Platform and reviewing test software. With this document, you will explore what you can do with the YTPC1002C01 platform.

∗ **If users are using this platform for the first time, please operate it as the following steps:**

1. Verifying the Kit Contents ---(Ch.3.1)
2. Installing PLUSDR Platform Hardware ---(Ch.3.2)
3. Computer Environment Setting ---(Ch.4)
4. Platform Software Quick Start ---(Ch.5)
5. Platform Software APIs And Examples ---(Ch.6)
6. Troubleshooting ---(Ch.7)

3. Verifying the Kit Contents

3.1 Kit Contents



1. YTPC1002C01 platform
2. DC Power adapter 12V with 110V / 220V power cord
3. 1G Ethernet - Cat 5e/Cat6 cable
4. SMA (Male)-to-SMA (Male) cable 2 pieces (option)

- **Preparing the Environment**

Ensure that the environment users are using the YTPC1002C01 platform in meets the following specifications.

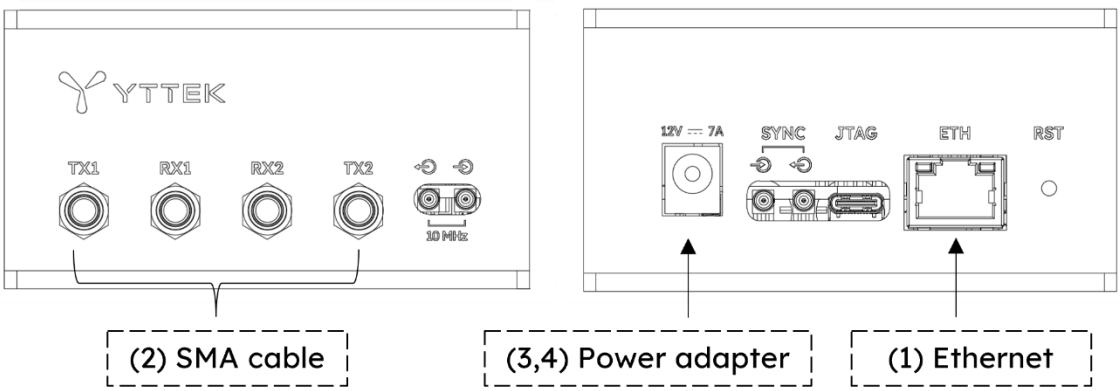
Ambient temperature range	0 °C to 45 °C
Optimized Operating temperature	23 °C ± 5 °C
Operating humidity	20% to 80% relative humidity, noncondensing
Indoor use only.	The platform is not waterproof and dustproof

⚠ Do not operate the YTPC1002C01 platform in an environment not specified as above. Product misuse can result in a hazard.

⚠ If the product is damaged, return it to YTTEK for repair. It is not recommended to repair by yourself.

3.2 Installing YTPC1002C01 Platform

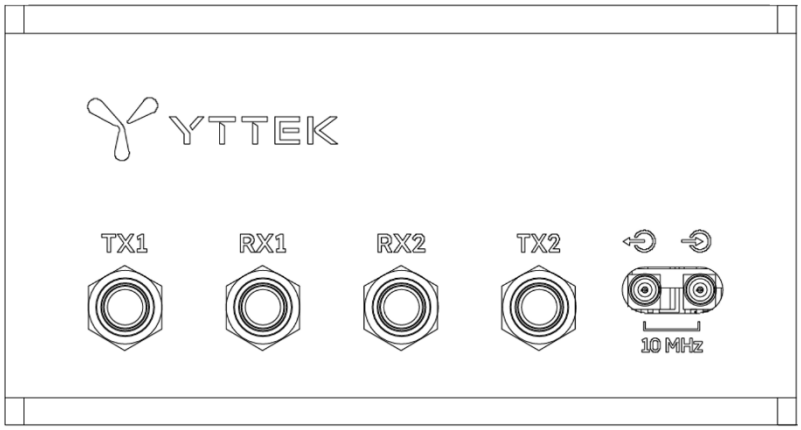
1. Connect the YTPC1002C01 platform to the computer by inserting an Ethernet cable into the “ETH” port. Kindly refrain from disconnecting the cable while the operation is in progress.
2. Connect SMA cables to confirm signal source.
3. Connect the 110V / 220V AC to 12V DC power adapter to the YTPC1002C01 platform. Please note that user need to use the power adapter provided by YTTEK to avoid any power issues.
4. Insert the 110V / 220V power cord of power adapter into a wall outlet and turn on platform.



- **Uninstalling YTPC1002C01 Platform**

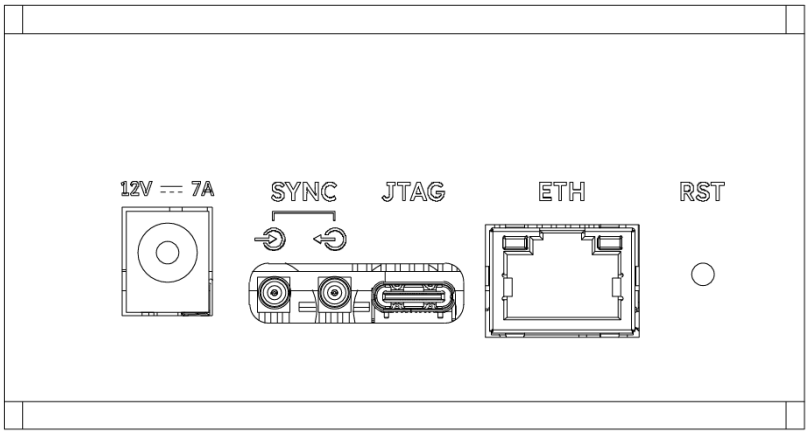
1. Unplug the 110V / 220V power cord and turn off platform.
2. Disconnect the 110V / 220V AC to 12V DC power adapter.
3. Unplug the Ethernet cable/SMA cables.

3.3 YTPC1002C01 Platform Front Panel



Connector		Use
SMA	Tx1	Input or output terminal for the RF signal on channel 1 and 2. ● female SMA connector ● 50 Ω impedance ● single-ended
	Rx1	
	TX2	
	Rx2	
10MHz		10MHz reference clock input/output for synchronizing with other equipments.

3.4 YTPC1002C01 Platform Back Panel



Connector	Use
DC IN	Input the 12V DC power by power adapter in the YTPC1002C01 kit contents.
ETH	Connect to computer for transaction of data and control signal. This port does not support 10M/100M Ethernet.
JTAG	A JTAG interface that connects the host computer to the device FPGA for update purposes. This port can be used with the Xilinx configuration tool to load a new bitfile provided by YTPC1002C01 platform.
SYNC	Input/output port to accept/provide a synchronization signal from/to another compatible device. This also acts as trigger input port in trigger mode.

4. Configuring the Device and Computer Environment Setting

4.1 Required Software

The software used to prepare this document is listed in the following:

- MATLAB R2018a ~ R2023a
- Windows 10 (Win 64-bit)

4.2 Ethernet Interface

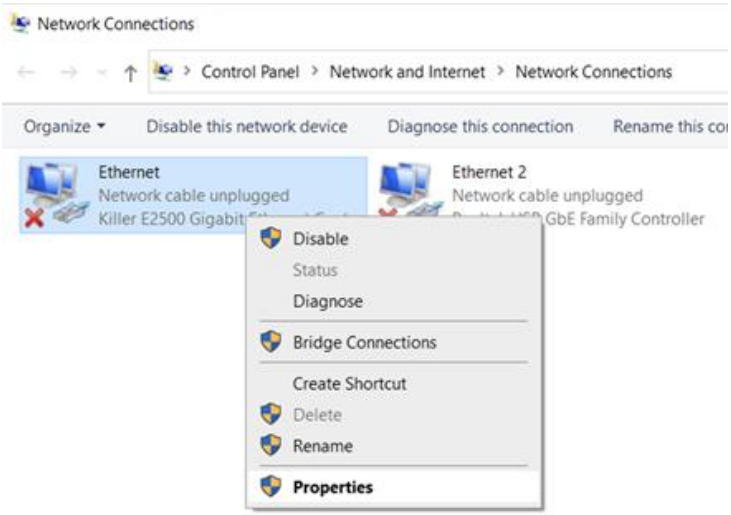
- Host Ethernet Interface—Single Device Configuration

Description	IP Address
Host Ethernet interface	192.168.3.200
Default Device IP address	192.168.3.100

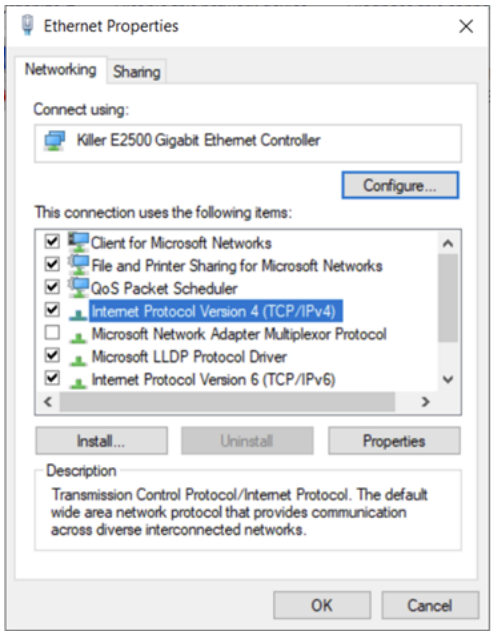
4.3 Configuring the Host Ethernet Interface

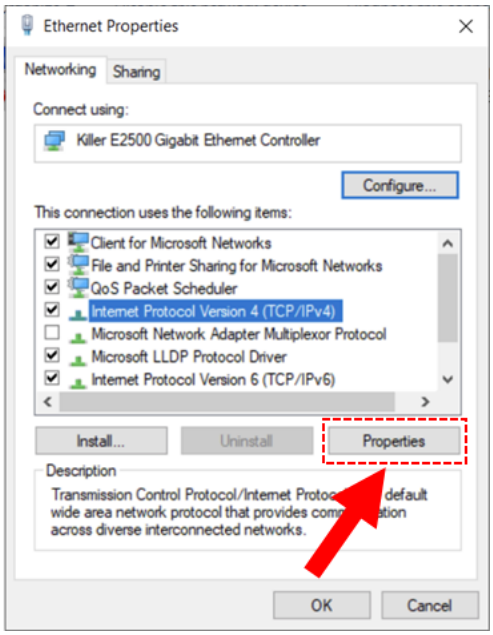
Configure the host Ethernet interface with a static IP address and set the IP address **"192.168.3.200"** on the host computer. If users use Ethernet adapter, please use "USB 3.0 to gigabit Ethernet adapter". Following the steps:

i. Choose the network adapter which is connected to the platform. Right click on the network adapter and choose "Properties".

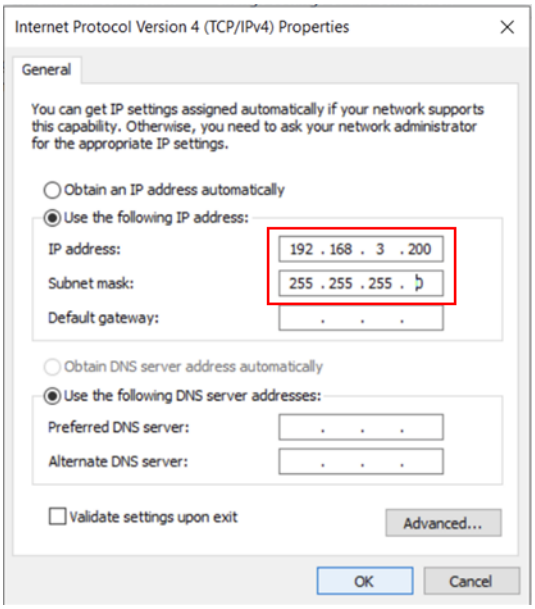


ii. Choosing the Internet Protocol Version 4 (TCP/IP). It is better to cancel all of the ticks in the menu except Internet Protocol Version 4 (TCP/IP).

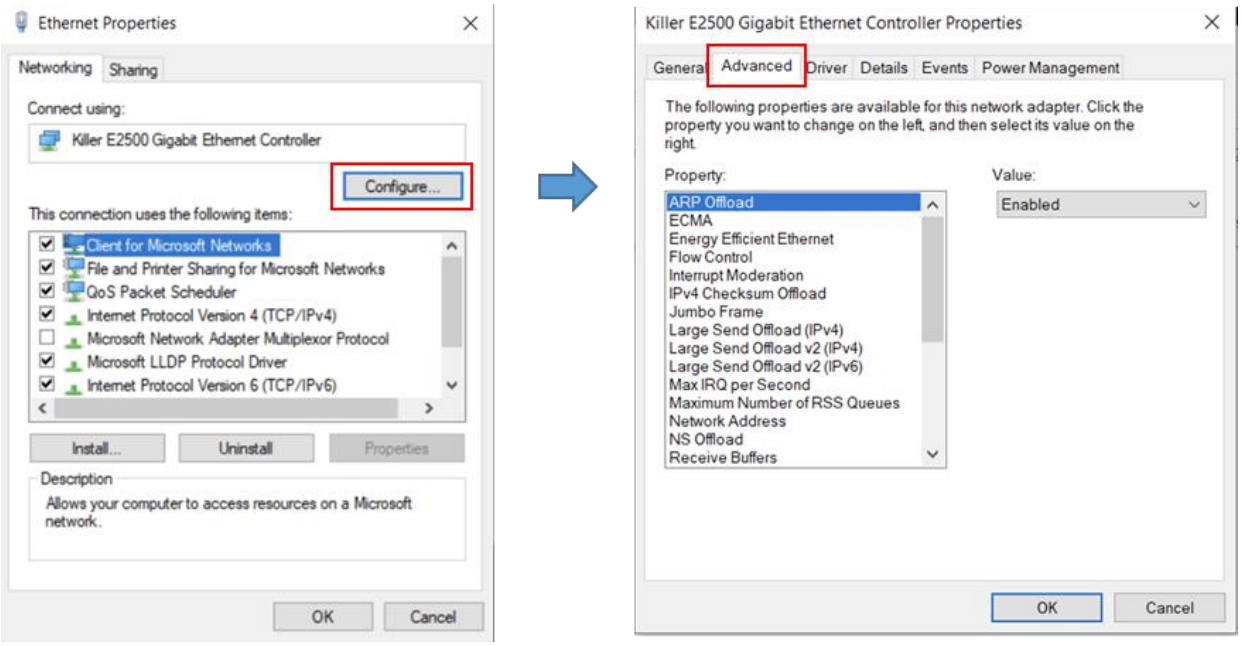




iii. Tick the label “using the following IP Address”. Fill the IP Address with 192.168.3.200, and subnet mask with 255.255.255.0., click OK and finish IP setting process.

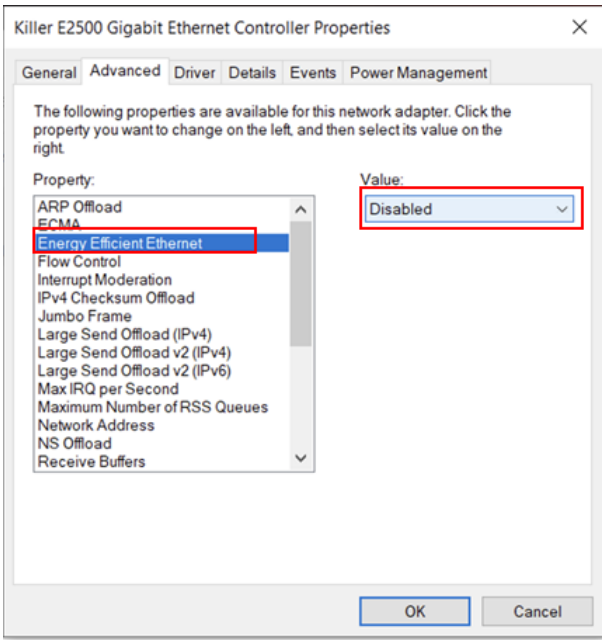


iv. Click the button “Configure…”, and select “Advanced” menu, change the value to turn off Ethernet energy-saving status. For the high-speed transmission, please disable “Energy-saving Ethernet”, click OK to finish Networking setting process.



Disabled

Power Saving Mode & Energy-saving Ethernet

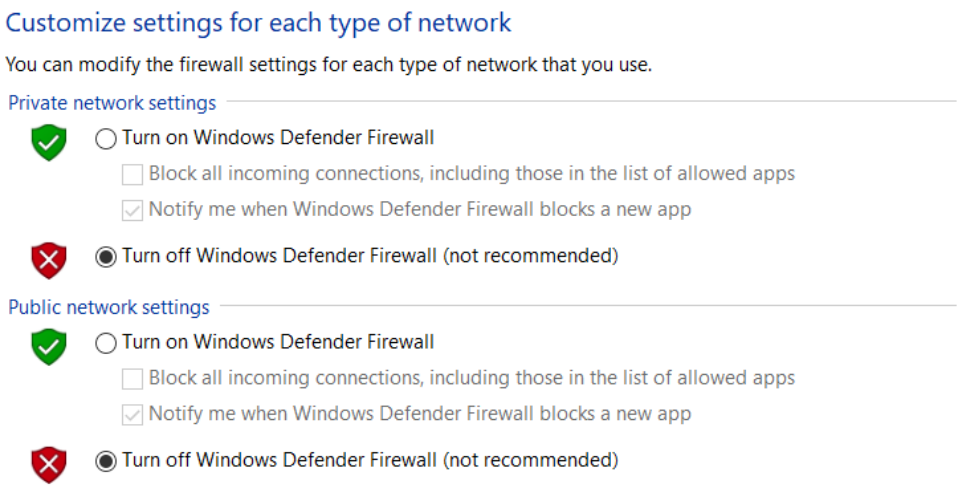


∗Note: Update ethernet driver to the latest version is recommend.

∗Note: If users need to use an USB to Ethernet adapter, YTTEK recommends using Realtek or Intel USB 3.0 to Gigabit Ethernet Adapter.

∗Note: If users use Ethernet adapter, please use "USB 3.0 to gigabit Ethernet adapter" and connect USB3.0 on the PC side. If users use USB 2.0, the transmission speed will be reduced, and the Ethernet packets may be lost.

v. Turn off the firewall to prevent the firewall from blocking network packets.



vi. If user still have problems with the network connection according to the above steps, please turn off antivirus software. And confirm if there is any software blocking the network packets.

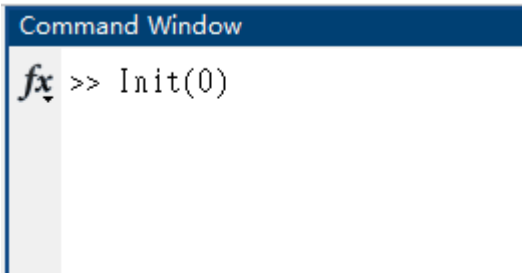
5. Platform Software Quick Start

When the platform power is turned on and the network connection is confirmed, users can enter the folder: " **.\release** " to initialize the platform. (The detail introduction of APIs in Chapter6.)

5.1 Step1: Initialization

First, enter "Init(type)" in the command window to initialize the platform.

- Type → 0: all platform are single, 1: Master slave mode. (If type is not entered, the default is 0.)



Please note that during the initialization of the YTPC1002C01 devices, ensure that no external input signals contain negative voltages, as these can interfere with the operation of internal components and potentially cause the initialization to fail.

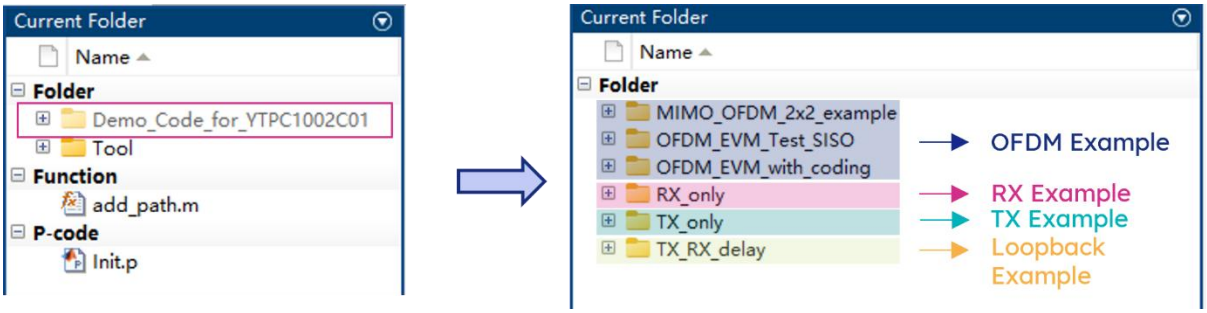
If the initialization is complete, the console would show the message "Software version:x.x.xxxx, HW Version: xx" and "Initial Completed! ".

The folder "Tool" contains all APIs for users. The related path of "Tool" will be added after platform is initialized.

```
>> Init
==========================================================================================
!!! Before system initialization, Please check one LED is ON only in each SDR hardware
Now, detecting exist SDR hardware ! Please wait...
==========================================================================================
 Software version: 2.10.2456710

 SDR idx 0: HW version: 8.3
 Initial active HW OK !!!
 clock set to 1500(mv), sdr 0
 Set Tx Attenuation OK !!!
 Set Rx Gain OK !!!
 ==========================================================
  Initial Completed !
 ==========================================================
```

5.2 Step2: Run Example

When the initialization is complete, users can choose the example to test. The detailed example is described in the next section. YTPC1002C01 examples are divided into two categories: basic_function and communication_example

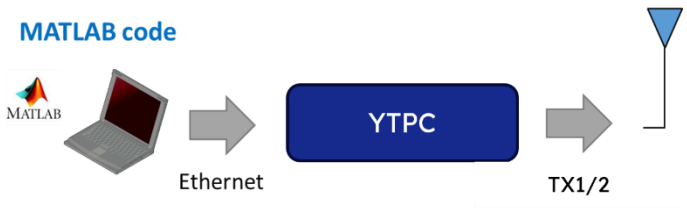


6. Introduction of MATLAB APIs and Examples

This section will introduce all the APIs of YTPC1002C01 platform. **All APIs are in the folder ".\release\tool".**
User can easily do the communication applications by using the APIs on YTPC1002C01 platform. This section will introduce some examples of communication applications. User can quickly understand how to build user's own algorithm for communication systems based on our examples.
All the example code are in the folder: ".\release\Demo_Code_for_YTPC1002C01".

6.1 Transmit I/Q raw data : TX()

- **TX(data, repetition, wait_time)**
  (1) Use Case1: TX(data, times)
  (2) Use Case2: TX(data, rep, seconds)
- Example path: " .\release\Demo_Code_for_YTPC1002C01\TX_only\TX_exp_with_TX_gain.m "



User could generate any I/Q raw data in MATLAB and transmits these I/Q raw data by calling the function TX(). The TX() function has three arguments "TX(xk, rep, time)", the first argument "xk" is the MATLAB complex variable and xk can be replaced with any other variable.

$$\mathrm{xk} \rightarrow \underbrace{\begin{bmatrix} 0.2+i0.5 & \cdots & 0.9-i0.1 \\ -0.3+i0.7 & \cdots & -0.2-i2 \end{bmatrix}}_{\text{Number of samples}} \Big\} \text{ Dimension of RF chain}$$

The row dimension means the number of TX RF chain, the first-row data will be transmitted to TX1, second-row data will be transmitted to TX2 and so on. The column dimension means the number of samples of the I/Q raw data. The limintation of total number of elements of xk is less than 8e7 (numel(xk) <= 8e7). If the SDR has 2 TX RF

chain, the limitation of row dimension is less than 2. The type of each element of xk should be complex and double for real and double for imagery part. User can use "whos" to see the attribute of xk.

```
>> whos xk
  Name      Size            Bytes  Class     Attributes

  xk        1x64             1024  double    complex
```

The range of the each real and imagery of each element of xk are $-1 \leq real(xk(m,n)) \leq 1$ and $-1 \leq imag(xk(m,n)) \leq 1$. If the value is exceed the range of ±1, we will clip to ±1. The time duration between $xk(m,n)$ and $xk(m,n+1)$ is the inverse of the sampling rate. In YTPC1002C01 the default sampling rate is 122.88MHz. For example, if user want to generate one period of single tone I/Q raw data with frequency 1.92MHz, user can generate like this

```
freq=1.92e6;
SR = 122.88e6;
ts = 1/(SR);
T = ceil(SR/freq);
xk = exp(1j*2*pi*freq*(0:ts:(1*T-1)*ts));
```

These codes are shown in the file "./Demo_Code_for_YTPC1002C01/TX_only/single_tone_generator.m"
xk will be a 1x64 complex array. If you set rep=1, YTPC1002C01 will transmit these 64 samples once then stop. If you set rep=2, YTPC1002C01 will tramsit thest 64 samples twice. If you want to transmit these 64 samples for a while. You can set rep=0 and set the third argument "time=10", YTPC1002C01 will transmit repeatedly these 64 samples last 10 senconds then stop. So, the second argument "rep" means how many times for transmitting xk if rep >0. "rep" should be positive integer. The third argument "time" means how long you want to tranmit xk, but rep should be set zero rep=0, the unit is second. If you set rep=0 nad time=1000, YTPC1002C01 will transmit I/Q raw data last 1000 seconds.

Case1: TX starts to transmit for once!

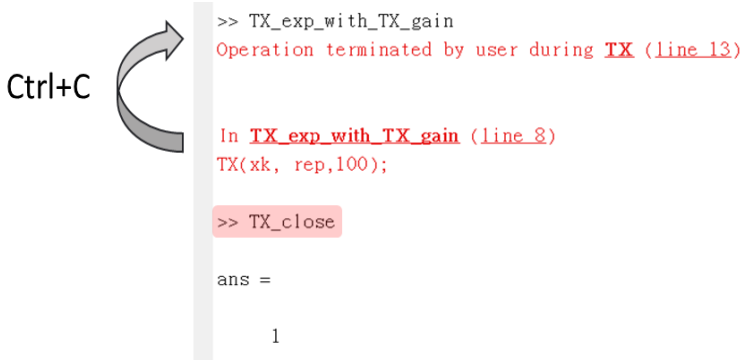
```
freq=1.92e6;
rep=0;  % infinite repeat
xk = single_tone_generator(freq);
TX(xk, 1);
```

Case2: TX starts to transmit for 10 seconds!

```
freq=1.92e6;
rep=0;  % infinite repeat
xk = single_tone_generator(freq);
TX(xk, rep,10);
```

If you want to stop transmitting before the time you set, press "Ctrl+c" and type

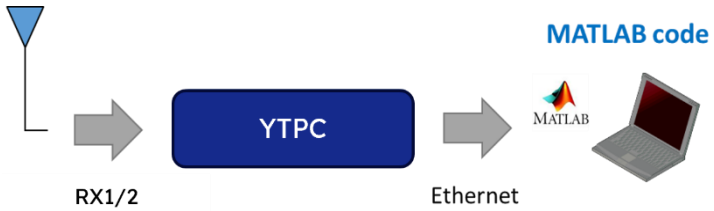
TX_close() then YTPC1002C01 will stop transmitting any thing.



6.2 Receive I/Q data : RX()

- **rx_data = RX(number_of_rows, rx_length)**
- Example path: " .\release\Demo_Code_for_YTPC1002C01\RX_only\RX_with_RX_gain.m "
- Log RF to basedband waveform

User can use this platform as a scope to easily log baseband I/Q data from RF signal into a variable in MATLAB. The maximum receive length is about 8e7 samples.

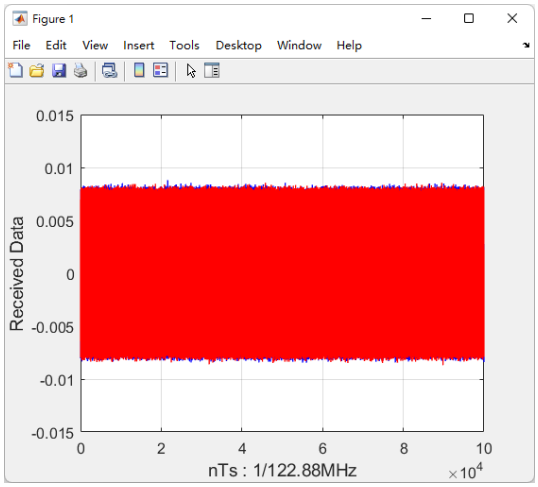


✧ Program: rx_len → log length

```
Number_of_RX_rows=2;  % 1:SISO, 2:MIMO
rx_len=100000;
set_RX_Gain([0 15; 1 15]);
rk = RX(Number_of_RX_rows, rx_len);
```

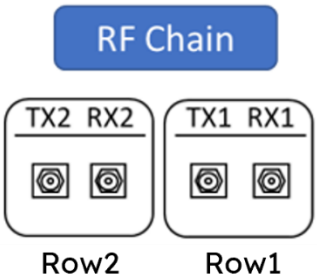
rk is the variable of the I/Q data. The maximum length of rk is about 8e7 samples, which depends on the size of RAM of user's computer. User can check the time domain waveform and frequency domain power spectrum density (PSD). Here show the measuring results as follows.

✧ Plot Result:



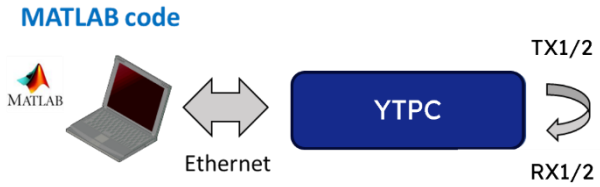
The number of rows in data to determine how many antennas to use to receive signals. Two parameters are needed in this function: the number of rows in data and receive length. There are two values can be chosen for the number of rows: 1 and 2. And so on, 1 is used for one antenna and 2 is used for two antennas in one YTPC1002C01. The parameter "length" is to determine how long the data would be received in a YTPC1002C01 platform. For example,

- ✧ Number of rows → 1: RX1
  - RX(1, 1e6): received 1 million samples for single antenna in the YTPC1002C01.
- ✧ Number of rows → 2: RX1&RX2
  - RX(2, 1e6): received 2 million samples for two antennas in the YTPC1002C01.



6.3 TRX Loop Back : TX_RX()

- **rx_data = TX_RX(TX_data, TX_rep, RX_length, RX_delay, LB_mode)**
- **This function is limited to a single platform.**
- **Feature: TX+RX** (*TX and RX center frequencies can be different.)
  TX_rep: TX transmit data once or repeatedly (repetition: 0, once: 1, twice: 2, and so on…)
  RX_length: RX log length
  RX_delay: RX delayed reception length
  LB_mode: 0: External loop back, 1: Internal loop back



This function allows the user to decide to transmit data once or repeatedly, and receive RX of any length and delay in the same time.

- **TX_data** has the same form as TX(), the row dimension of tx_data means the number of TX RF chain and the first-row data will be transmitted to TX1, second-row data will be transmitted to TX2 and so on. If the SDR has 2 TX RF chain, the limitation of row dimension is less than 2. The type of each element of xk should be complex and double for real and double for imagery part.
- **TX_rep** defines how many times TX_data can be transmitted.
- **RX_delay** can control the receiver to delay the reception by a few sampling points, and the user can discard unnecessary data directly.
- **LB_mode** can choose external or internal loopback. Internal loopback mode means TRX exchange data used by FPGA DDR memory. External loopback mode means TRX exchange data used by RF transmission.

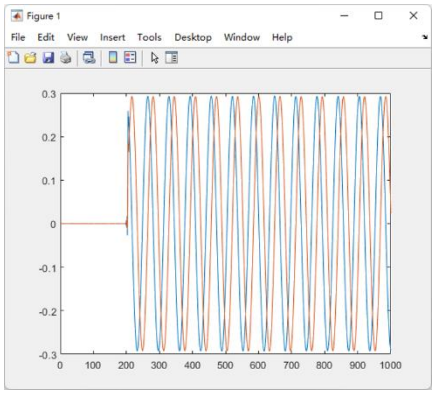
6.3.1 Example 1: Single Tone Loopback Test

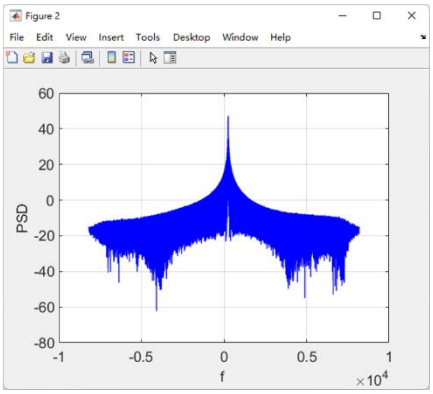
- Ensure that cables or antennas are connected to the YTPC1002C01 TX1-RX1.
- path:
  " .\release\Demo_Code_for_YTPC1002C01\TX_RX_delay\Test_TX_RX_delay_SISO.m "

✧ Program:

```
TX_REP=0; % 0: infinite repeat;   1: once
RX_LEN=1000;
RX_delay=0;
LB_mode=0; % 0: RF; 1:memory loop back

rec_data=TX_RX(tx_data*0.8, TX_REP, RX_LEN, RX_delay, LB_mode);
```

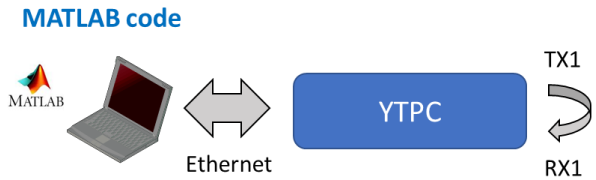
✧ **Plot Result:**





6.3.2 Example 2: OFDM Loop Back Test

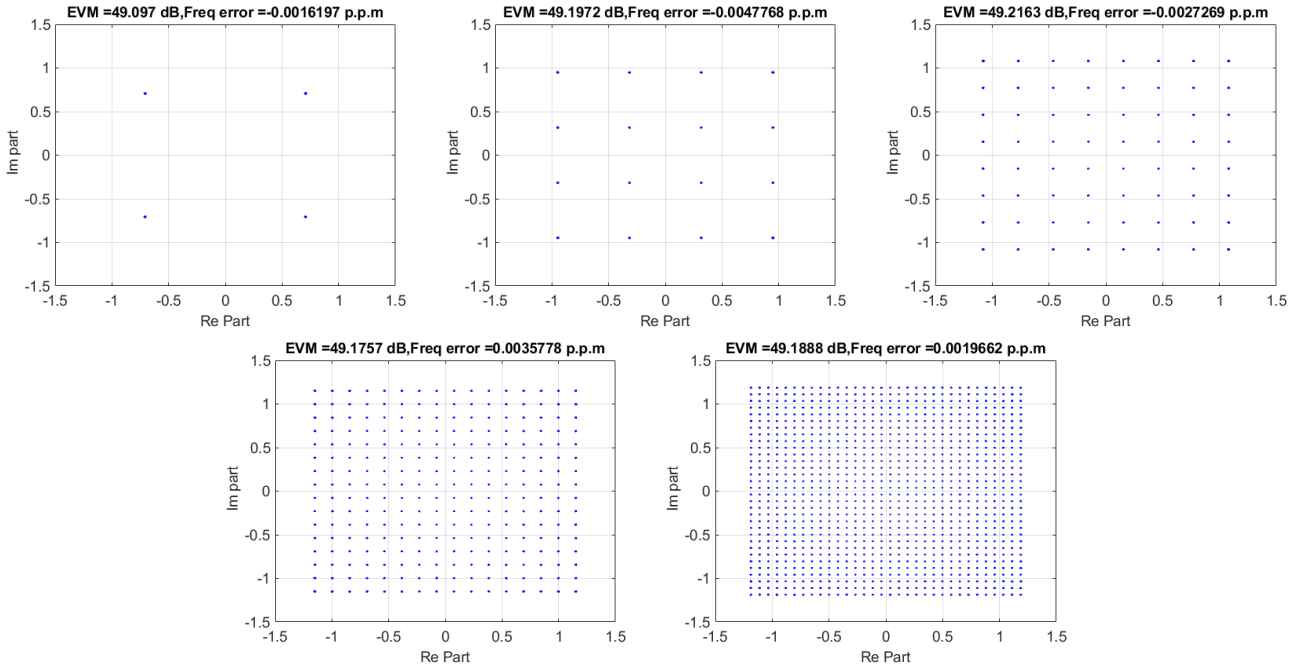
- This example familiarizes users with the OFDM transmission on the YTPC1002C01 platform.
- Ensure that cables or antennas are connected to the YTPC1002C01 TX1-RX1.
- If user need to know the signal specifications, please refer to YTPC100x SISO_OFDM.pdf
- path:
  " .\release\Demo_Code_for_YTPC1002C01\OFDM_EVM_Test_SISO\Test_EVM_TXLB.m "



✧ Program:

```
N_cpc_data=(mod(idx_test,5)+1)*2;
C_init_data=1253233+idx_test_fixed*1000;
sn=TX_RJ_v4(Fs, Inx_Used_Carrier, total_num_sym, FFT_SIZE, CP_len, N_cpc_data, C_init_data, C_init_RS, preamble_seq_all);

RX_LEN=length(sn)+1000;
rn=TX_RX(sn*0.4, TX_REP, RX_LEN, RX_delay, LB_mode);
```

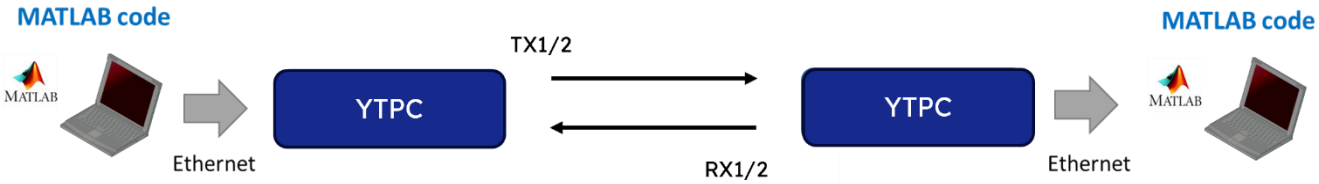
✧ Plot Result: QPSK/16QAM/64QAM/256QAM/1024QAM



6.4 OFDM Example

6.4.1 Example 3: OFDM EVM Test with Coding (Transfer Pictures)

- Transpose data to picture based on OFDM_EVM_Test_SISO and do the encoding processing.
- PC1(TX) path:
  " .\release\Demo_Code_for_YTPC1002C01\OFDM_EVM_with_coding\Picture_with_coding_BS.m "
- PC2(RX) path:
  " .\release\Demo_Code_for_YTPC1002C01\OFDM_EVM_with_coding\Picture_with_coding_UE.m "



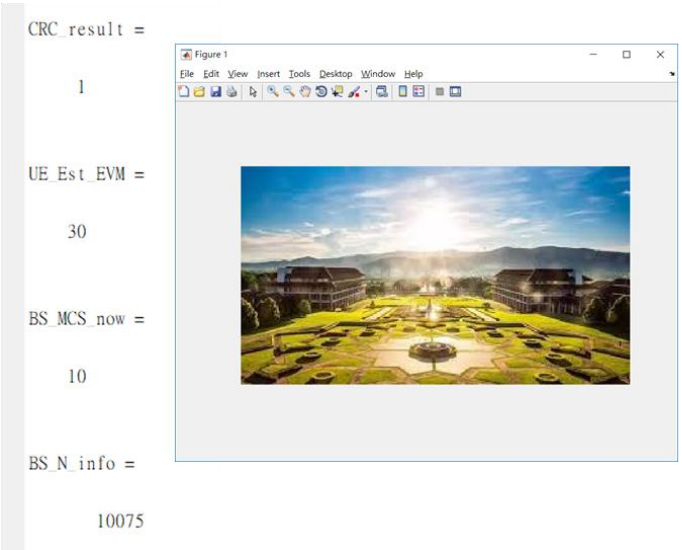
BS

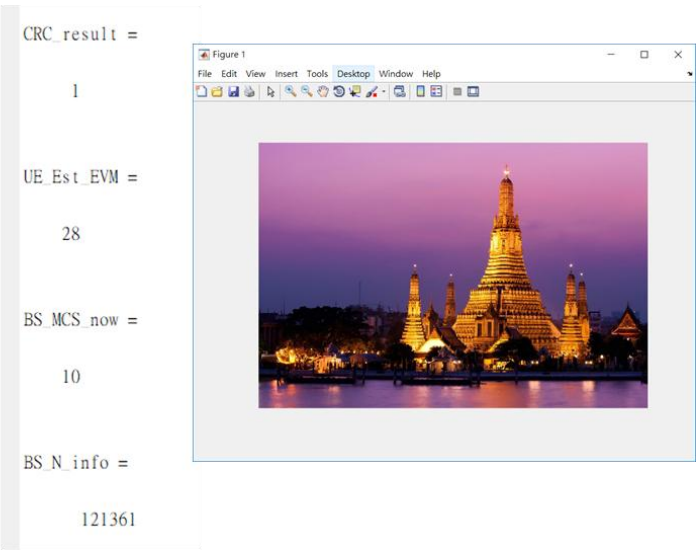
```
%=== open picture file and generate information bits ========
fid = fopen(strcat(name,'_', num2str(seq_image,'%05d'), '.jpg'), 'r');
A = fread(fid, 'uint8');
B = de2bi(A);
C = reshape(B', 1, numel(B));
BS.TX.info_ak=C;
seq_image=seq_image+1;
seq_image=mod(seq_image,seq_image_max)+1;
fclose(fid);
%==========================================================

sn=BS_TX_state14_v1(Inx_Used_Carrier, FFT_SIZE, CP_len,Inx_sym_preamble,preamble_seq);
TX(sn*0.4, 1);
```

UE

```
rn = RX(1,2000000*2);
CRC_result=UE_RX_ctrl_r2(rn,Inx_Used_Carrier, FFT_SIZE, CP_len,Inx_sym_preamble,preamble_seq);
```

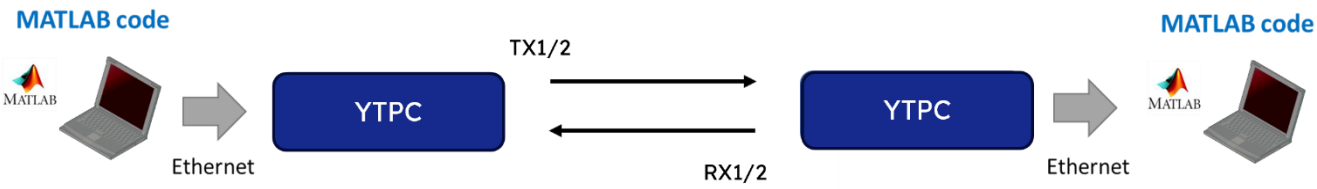
Result for UE





6.4.2 Example 4: MIMO OFDM Test

- Here is an example to let users understand how to do 2x2 MIMO transmission with the YTPC1002C01 platform. This example requires TRX1 and TRX2
- If user needs to know the signal specifications, please refer to YTPC100 SDR_MIMO_OFDM.pdf
- PC1(TX) path:
  " .\release\ Demo_Code_for_YTPC1002C01\MIMO_OFDM_2x2_example\ Test_EVM_Tx_MIMO_2x2_two_layer_XP.m "
- PC2(RX) path:
  " .\release\ Demo_Code_for_YTPC1002C01\MIMO_OFDM_2x2_example\ Test_EVM_Rx_MIMO_2x2_two_layer_XP.m "



✧ Program:

TX

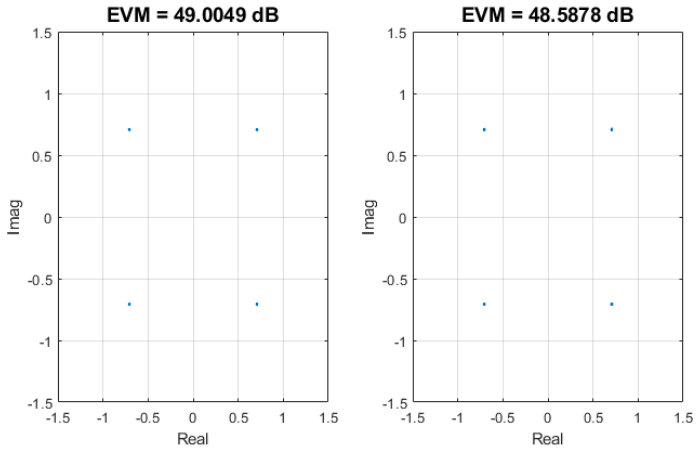
```
Data_Parameter.N_cpc_Data = (mod(idx_test,5)+1)*2;

%=== MIMO-OFDM TX generation ===
sn = gen_MIMO_OFDM_TX_r1_XP(System_Parameter.Num_Layer, Data_Parameter.N_cpc_Data, Data_Parameter.C_init_Data, Preamble_Parameter.preamble_seq_all);
%==============================

% TX
TX(sn*0.4, 1);
```

RX

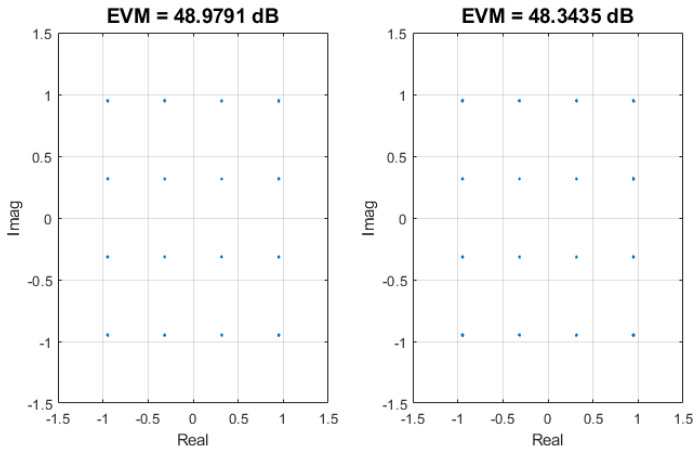
```
% RX
rn = RX(2,224128*2 + 1000);

[N_cpc_est, inx_start]=find_signal_header_CF(rn(1,:), System_Parameter.Inx_Used_Carrier,System_Parameter.FFT_SIZE,System_Parameter.CP_Len);
[rn_Equal, rn_Equal_Data, rn_Equal_Data1, rn_Equal_Data2, EVM, EVM1, EVM2, fe_est] = detect_MIMO_OFDM_RX_r2_XP(rn, 0, ...
                                                                                    System_Parameter.Num_Layer,  Data_Parameter.C_init_Data, ...
                                                                                    Preamble_Parameter.preamble_seq_all, N_cpc_est, inx_start);
```

✧ Plot Result:

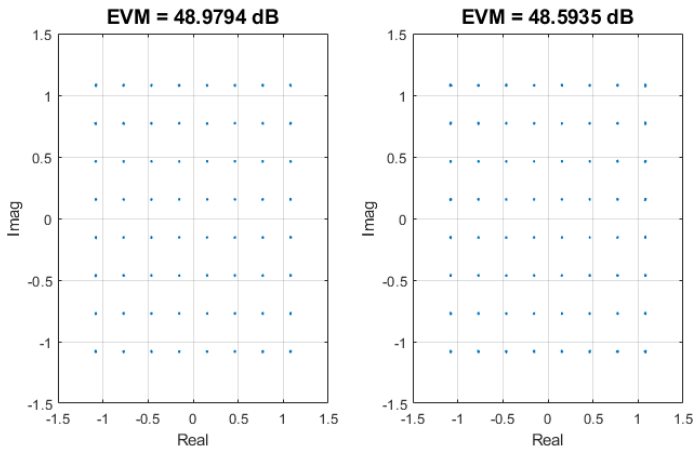
■ QPSK



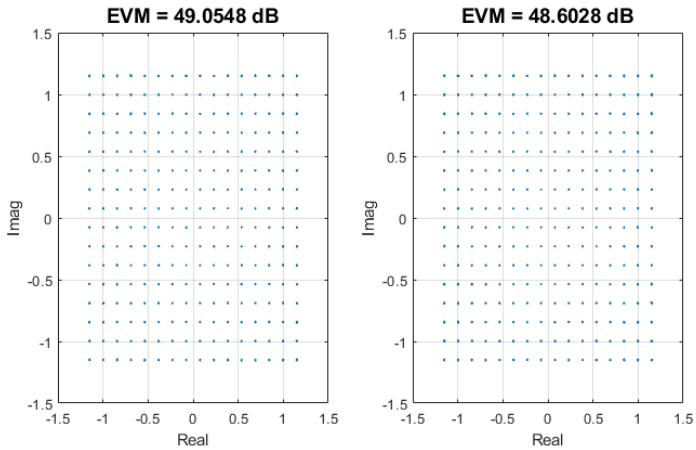
■ 16QAM



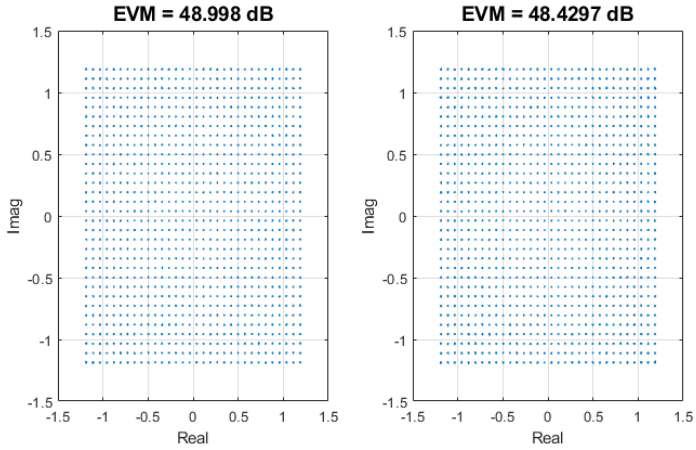
■ 64QAM



■ 256QAM



■ 1024QAM



6.5 RF Setting

Before executing "**set_TX_power()**" and "**set_RX_Ref_Level_ELSDR()**", users must ensure that the LO is set to the target frequency. We recommend executing "**LO_Change()**" prior to adjusting the RF settings to prevent errors during configuration.

6.5.1 Set Transmitted Power: set_TX_power()

- **set_TX_power([RF_chain0 Amp_val0; RF_chain1 Amp_val1])**
- Input array: ([RF_chain0 Amp_val0; RF_chain1 Amp_val1])
- For example: set Tx power -10dBm:
  set_TX_power([0 -10; 1 -10]); % dBm, set channel power

This type is used to control the transmitted power by TX amplitude(dBm). This function will modify the TX power by using the calibration table. YTTEK provides this API to allow users to directly set TX amplitude(dBm).

6.5.2 Set RX Reference Level: set_RX_Ref_Level_ELSDR()

- **set_RX_Ref_Level_ELSDR([RF_chain0 Amp_val0; RF_chain1 Amp_val1])**
- Input array: ([RF_chain0 Amp_val0; RF_chain1 Amp_val1])
- Level Range: +10 ~ -40 dBm
- For example: set Rx Reference Level -5dBm:
  set_RX_Ref_Level_ELSDR([0 -5; 1 -5]); % dBm, set channel power

This function is used to control the receive gain via RX reference level (dBm). Reference level is the top edge of display and represents the maximum expected power at the YTPC1002C01 RX input.

6.5.3 Set Center Frequency : LO_CHANGE()

- **LO_CHANGE(idx_SDR, tx_lo_1, rx_lo_1)**
- SDR default frequency: 1GHz
- Frequency range: 300MHz~6GHz, index SDR: 0
- Unit: MHz
- There is one LO in each platform. TRX1&TRX2 share the LO.

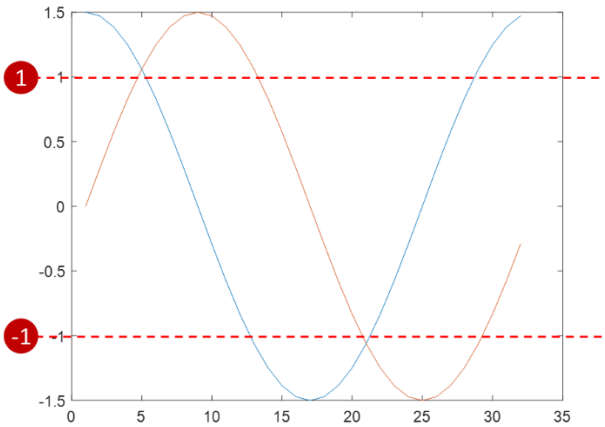
This API is used for changing center frequency of SDR. Three parameters are needed in this function, index of SDR, TX Frequency, RX Frequency.

✧ For example: SDR:0, TRX1&TRX2 Frequency: 4GHz.

```
>> LO_CHANGE(0,4000,4000)
```

7. Troubleshooting

- About TX
  1. Input TX signal amputitude can't exceed ±1.



  2. Input TX signal should be a complex row vector.
  3. The maximum number of samples in TX signal is 8e7 samples in total (2TX).
- About APIs
  1. Please ensure the TX attenuation and RX gain on any communication to avoid signal saturation due to the too low attenuation or too high gain.
  2. Calibration IQ Imbalance is limited to Loopback situations.

8. Document Version History

Version	Description	Author	Data	
V1.0	Release on June, 2025 Modify file description details	Terry @ YTTEK	2025/06/10	
V1.1	Modify API description	Terry @ YTTEK	2025/10/22	